

Change Risk Forecasting & Outage Prediction using Agentic AI - Proposal

Edward Sekharan
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| Theme

Innovation in SRE & Observability using Agentic AI

In today's fast-moving engineering environments, maintaining reliability while innovating rapidly is a constant challenge. Our hackathon entry embraces the theme of "Innovation in SRE & Observability using Agentic AI" by leveraging autonomous AI agents to bring a new level of proactivity and transparency to change management.

| Hackathon Topic

Change Risk Forecasting & Outage Prediction using Agentic AI

We propose an AI-driven workflow that analyzes every incoming change request—be it code, infrastructure, or configuration—and predicts its risk of causing service degradation or outages. Beyond a simple score, our system offers clear reasoning, actionable recommendations, and an explainability dashboard to guide teams through safe deployments.

| Industry Challenge

High-velocity engineering teams deploy changes constantly, yet each deployment carries the possibility of introducing performance regressions, misconfigurations, or full-blown outages. Traditional rule-based risk assessments are brittle: they miss contextual nuances, fail to learn from historical incidents, and lack the explainability teams need to trust automated recommendations.

The result is either overly conservative processes that slow innovation or blind spots that lead to costly downtime.

| Proposed Solution Approach

How can we build an autonomous AI agent that analyzes a change request and forecasts its risk level intelligently—complete with reasoning, recommendations, and explainability—before the change is deployed?

1

Intelligent Change Ingestion

- **Natural-Language Parsing:** Agent ingests the change description, justification, and planned timing.
- **Contextual Enrichment:** Ties the request to relevant configuration items (CIs) and past incidents via a knowledge graph.
- **Initial Risk Estimate:** Generates a "low/medium/high" risk label based on criteria such as business-criticality and deployment window.

2

Historical Contextual Analysis

- **Similarity Search:** Finds past changes with matching keywords or affected systems.
- **Incident Correlation:** Adjusts risk upward if similar changes had incidents, or downward if historical rollouts were smooth.
- **Data-Driven Reasoning:** Provides human-readable explanations like, "No prior incidents for this CI in the last 90 days — risk lowered."

3

Automated Testing Mitigation Validation

- **CI/CD Integration:** Checks for existence and results of pre-production tests.
- **Mitigation Checks:** Verifies rollback plans or feature flags are in place when tests can't run.
- **Dynamic Risk Adjustment:** Updates risk score and suggests next steps, e.g., "Proceed with caution; ensure rollback script is validated."

4

Scheduling Conflict Detection

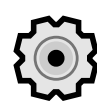
- **Calendar Scanning:** Detects overlapping changes on shared resources.
- **Team Coordination:** Alerts relevant teams via chatops or email when conflicts arise.
- **Recommendation:** Proposes rescheduling windows to minimize collision.

5

Final Recommendation & Explainability

- **Risk Level:** Delivers a final green/yellow/red verdict.
- **Actionable Guidance:** Provides a concise summary, e.g., "Yellow — deploy after verifying load-test results; rollback ready."
- **Explainability Dashboard:** Visualizes contributing factors, historical data points, and step-by-step reasoning for stakeholder review.

Key Features & Benefits



Agentic Autonomy

Minimal manual tuning—our AI agent learns and improves as more change outcomes are ingested.



Full Transparency

Every recommendation is backed by a clear explanation, boosting trust and adoption.



Reduced Outages

By catching high-risk changes early, we expect a significant drop in post-deployment incidents.



Scalable Extensible

Modular design allows plugging in new data sources (e.g., performance metrics, error budgets).

Technical Stack

AI Agent Framework

ServiceNow AI Agent Studio (or similar orchestration tooling)

Natural Language & ML

Transformer models for text understanding, historical risk models for statistical insights

Data Fabric & Knowledge Graph

Real-time access to change records, incidents, and CIs

Integrations

CI/CD pipelines, testing platforms, chatops (Slack/MS Teams), and monitoring systems

Frontend

Lightweight web dashboard for explainability and audit trails

Future Directions

- **Real-Time Observability Feeds:** Ingest live metrics to refine risk scores during the deployment window.
- **Cross-Project Learning:** Share anonymized learnings across teams to improve global model accuracy.
- **Self-Healing Automations:** Extend agents to automatically roll back risky changes when certain thresholds are breached.

Prepared for the "Innovation in SRE & Observability using Agentic AI" Hackathon

AI Agent Workflow for Change Risk Forecasting & Outage Prediction - Implementation

A ServiceNow AI Tools Analysis

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1. Introduction

Managing changes in an enterprise IT environment is a delicate balancing act. A single misstep can lead to outages, downtime, or unexpected risks that disrupt critical business operations. The "AI Agent Workflow for Change Risk Forecasting & Outage Prediction" addresses this challenge by harnessing ServiceNow's AI tools to automate the process, forecast risks, detect conflicts, and provide actionable insights.

This report walks you through the workflow step-by-step, showing how AI agents process change requests, predict potential issues, and ensure transparency with an explainability dashboard. By blending cutting-edge technology with a human-centered approach, this workflow helps teams make informed decisions, prevent disruptions, and continuously improve their change management processes.

2. Background on ServiceNow AI Tools

ServiceNow's AI capabilities are designed to streamline complex workflows like change management. Here's a quick look at the key tools powering this process:

- AI Agent Studio:** A platform for building custom AI agents with minimal coding, perfect for defining workflows and automating tasks.
- Now Assist:** A generative AI tool that interprets data, provides explanations, and offers recommendations in plain language.
- AI Agent Orchestrator:** Manages multiple AI agents, ensuring they work together smoothly across different steps.
- Knowledge Graph:** Connects change requests to configuration items (CIs), past changes, and incidents for richer context.
- Workflow Data Fabric:** Provides real-time access to all types of data, enabling quick historical analysis.
- Integration Hub:** Links ServiceNow to external systems, like testing tools, for seamless integration.
- ServiceNow Reporting:** Creates dashboards to visualize risks and recommendations for stakeholders.

These tools combine to form a robust system that not only predicts risks but also explains them, making the process transparent and trustworthy.

3. Detailed Workflow Breakdown

3.1 Input Collection and Initial Analysis

The journey begins when a change request lands in ServiceNow. The AI agent jumps into action:

What Happens: It pulls out key details—like the description, justification, and planned downtime—from the request.

Tools Used: Now Assist, AI Agent Studio, and Knowledge Graph.

How It Works: Now Assist reads the text and figures out what's being asked, while an AI agent built in AI Agent Studio ties the request to related systems using the Knowledge Graph. Then, it predicts the initial risk (low, medium, or high) based on factors like downtime timing or how critical the affected business group is.

Example: A change scheduled during peak hours might get flagged as "High risk due to potential business impact."

Output: An initial risk score with a clear explanation for the team to review.

3.2 Historical Analysis

Next, the AI looks back to learn from the past:

What Happens: It searches historical data for similar changes based on details like description or affected group.

Tools Used: AI Agent Studio, Workflow Data Fabric, and Machine Learning Models.

How It Works: Using Workflow Data Fabric, the agent digs into past records in real time. If it finds matches, it checks for related incidents. Successful past fixes lower the risk; unresolved issues raise it. If there's no history, it flags the change for extra scrutiny.

Example: "No incidents for similar changes, risk lowered to medium."

Output: An updated risk score with reasoning tied to historical insights.

3.3 Testing and Mitigation Validation

Before greenlighting a change, the AI ensures it's safe:

What Happens: It checks if testing in a pre-production environment is possible and reviews the results. If testing isn't an option, it evaluates backup plans.

Tools Used: AI Agent Studio, Integration Hub, and Now Assist.

How It Works: The agent uses Integration Hub to connect with testing tools and confirms if tests passed (lowering risk) or failed (raising it). If testing isn't feasible, it checks for solid mitigations—like a backup plan—and adjusts the risk accordingly.

Example: "Proceed with caution, ensure backup plan is ready."

Output: A risk assessment with practical next steps.

3.4 Conflict Detection

Timing is everything, so the AI scans for scheduling clashes:

What Happens: It looks for overlapping changes or shared resource conflicts.

Tools Used: AI Agent Studio and AI Agent Orchestrator.

How It Works: The agent checks the change schedule, and the Orchestrator coordinates across teams to spot issues. If a conflict pops up, it suggests rescheduling.

Example: "Conflict detected with another downtime, delay deployment."

Output: A conflict status with a recommendation to keep things smooth.

3.5 Final Recommendation

Now, it's decision time:

What Happens: The AI pulls together all the data to assign a final risk level—green (go), yellow (caution), or red (stop)—and gives advice.

Tools Used: AI Agent Orchestrator and Now Assist.

How It Works: The Orchestrator combines inputs from earlier steps, and Now Assist crafts a clear recommendation, like "Yellow: Proceed with a backup plan due to untested change."

Output: A final risk level with actionable guidance.

3.6 Post-Incident Analysis

If something goes wrong after deployment, the AI steps in to fix it:

What Happens: It links incidents to recent changes, finds the root cause, and suggests fixes.

Tools Used: AI Agent Studio and Now Assist.

How It Works: The agent matches incident details to changes, and Now Assist analyzes patterns from past resolutions to recommend solutions and prevent future issues.

Example: "Revert change X, apply patch Y to resolve."

Output: A detailed report with fixes and prevention tips.

3.7 Feedback Loop

Learning never stops:

What Happens: The AI gathers feedback on how changes went—incidents, performance, etc.—to sharpen its predictions.

Tools Used: AI Agent Studio and Machine Learning Models.

How It Works: Feedback refines the risk models, making the AI smarter over time.

Output: Improved accuracy for future forecasts.

3.8 Handling Edge Cases

The workflow stays flexible:

What Happens: For incomplete requests, it asks for more info or flags them as risky. Urgent changes can skip checks with manual approval and recorded. The urgent change will be fed back to the loop and the relevant mitigation steps are implemented, also for the knowledge base and model learning.

Tools Used: AI Agent Studio and Now Assist.

How It Works: The agent prompts users or adjusts risk dynamically, ensuring nothing slips through the cracks.

3.9 Explainability Dashboard

Transparency is key:

What Happens: A dashboard shows the risk score, factors like downtime or incidents, and recommendations.

Tools Used: ServiceNow Reporting and Now Assist.

How It Works: Reporting tools visualize the data, and Now Assist explains it in simple terms for everyone to understand.

Output: A clear, visual summary for stakeholders.

4. Summary of Tools and Roles

Here's a quick snapshot of how the tools fit into the workflow:

Step	Tools	Role
Input & Initial Analysis	Now Assist, AI Agent Studio, Knowledge Graph	Extract details and predict initial risk.
Historical Analysis	AI Agent Studio, Workflow Data Fabric, ML Models	Assess risks using past data.
Testing & Mitigation	AI Agent Studio, Integration Hub, Now Assist	Validate changes and mitigations.
Conflict Detection	AI Agent Studio, AI Agent Orchestrator	Spot scheduling conflicts.
Final Recommendation	AI Agent Orchestrator, Now Assist	Deliver a final risk level and advice.
Post-Incident Analysis	AI Agent Studio, Now Assist	Analyze incidents and suggest fixes.
Feedback Loop	AI Agent Studio, ML Models	Refine predictions with feedback.
Explainability Dashboard	ServiceNow Reporting, Now Assist	Show risks and reasoning visually.

5. Conclusion

This AI Agent Workflow transforms change management into a proactive, data-driven process. By configuring and training ServiceNow's AI tools, it forecasts risks, prevents outages, and keeps everyone in the loop with clear explanations. From analyzing past changes to suggesting fixes after incidents, it's a full-circle approach that gets smarter with every deployment.

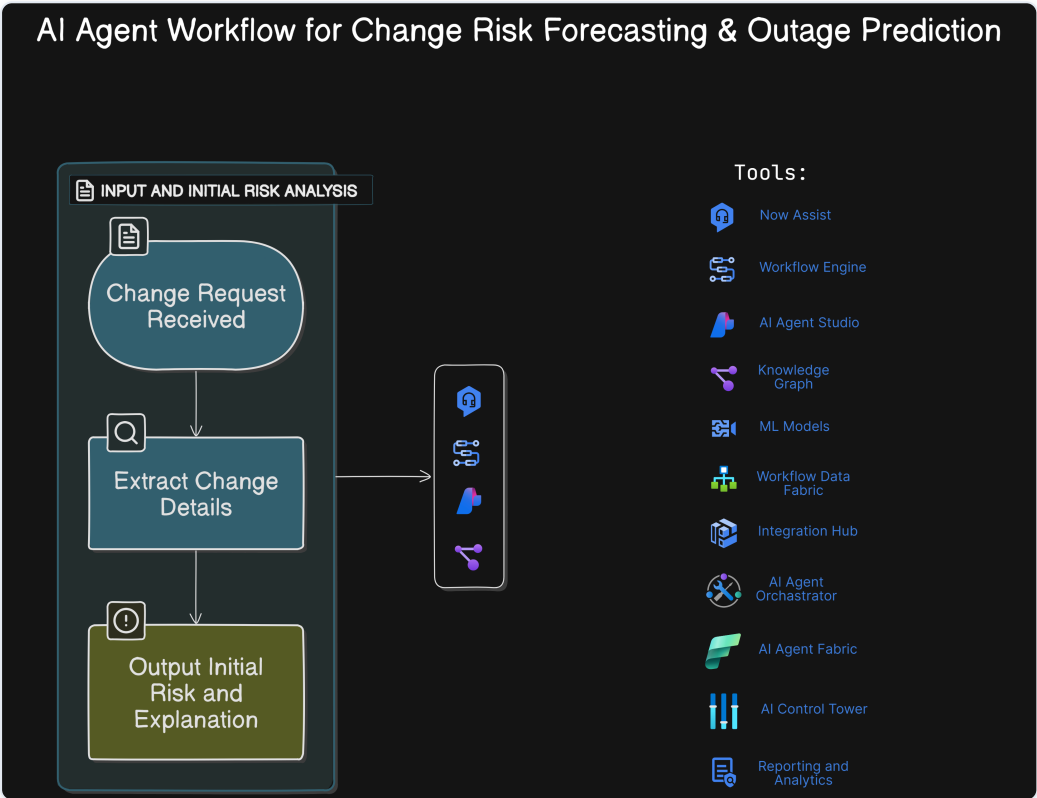
The result? Fewer disruptions, better decisions, and a more reliable IT environment—all powered by AI that's as practical as it is innovative.

AI Agent Workflow for Change Risk Forecasting & Outage Prediction

Visual Documentation & Work Flow

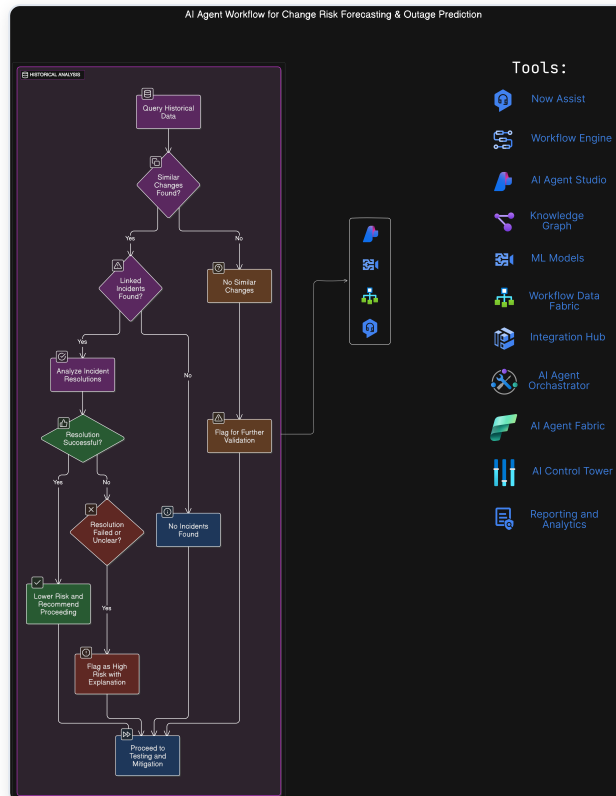
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Step 1: Input Collection & Initial Analysis



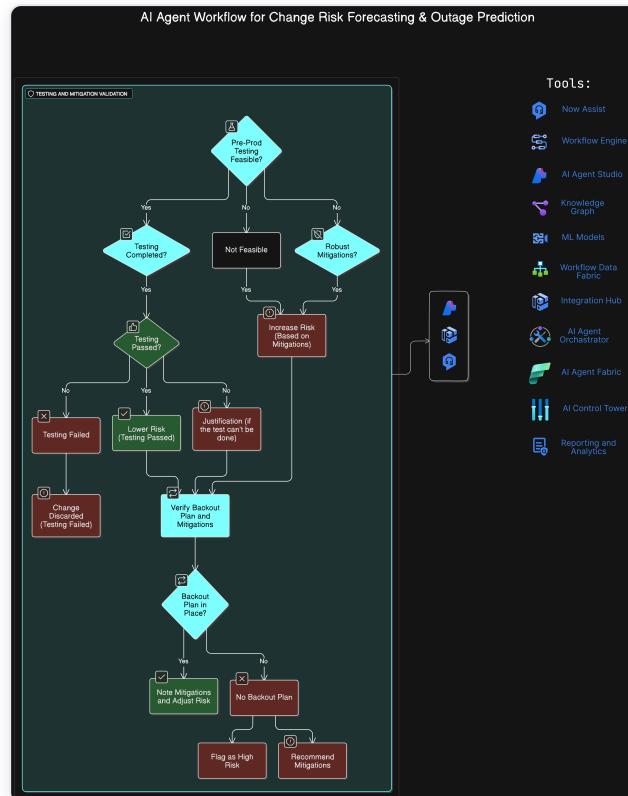
AI agent processes incoming change requests, extracts key details, and generates initial risk assessment

Step 2: Historical Contextual Analysis



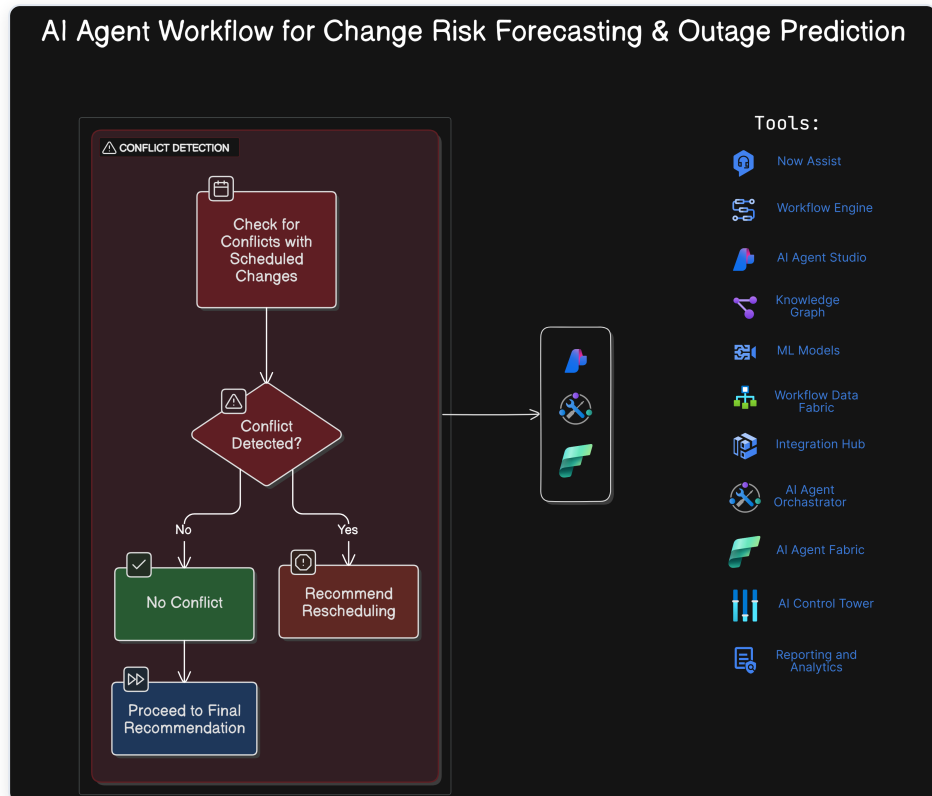
System analyzes historical data to find similar changes and correlate with past incidents

Step 3: Testing & Mitigation Validation



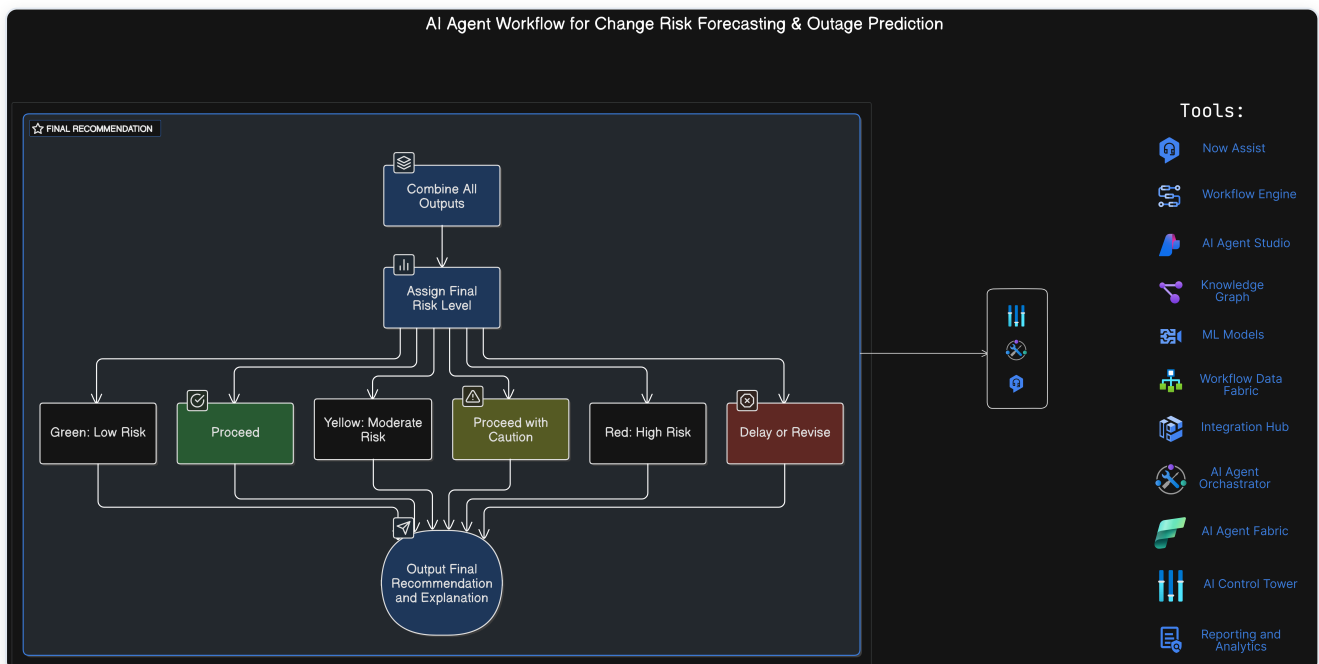
Validation of pre-production testing results and mitigation strategies

Step 4: Scheduling Conflict Detection



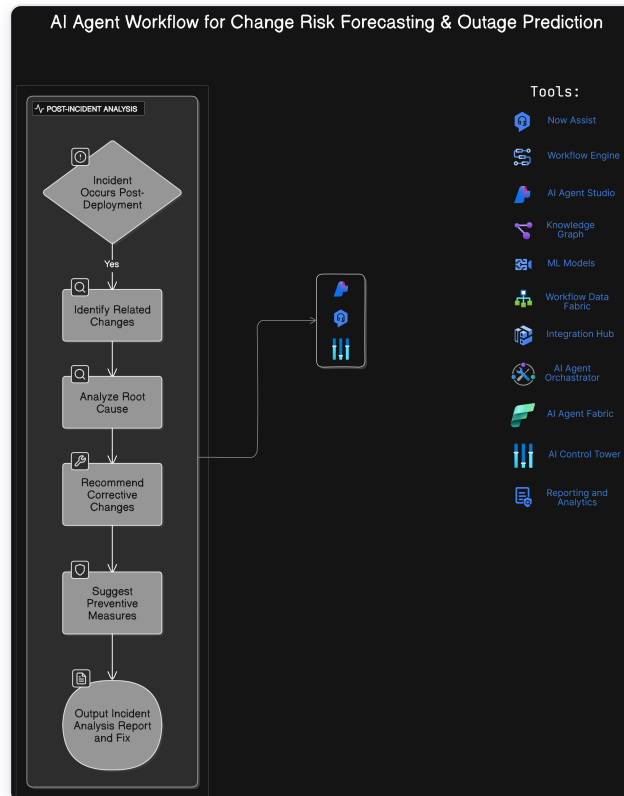
Detection of scheduling conflicts and resource contention across teams

Step 5: Final Recommendation & Decision



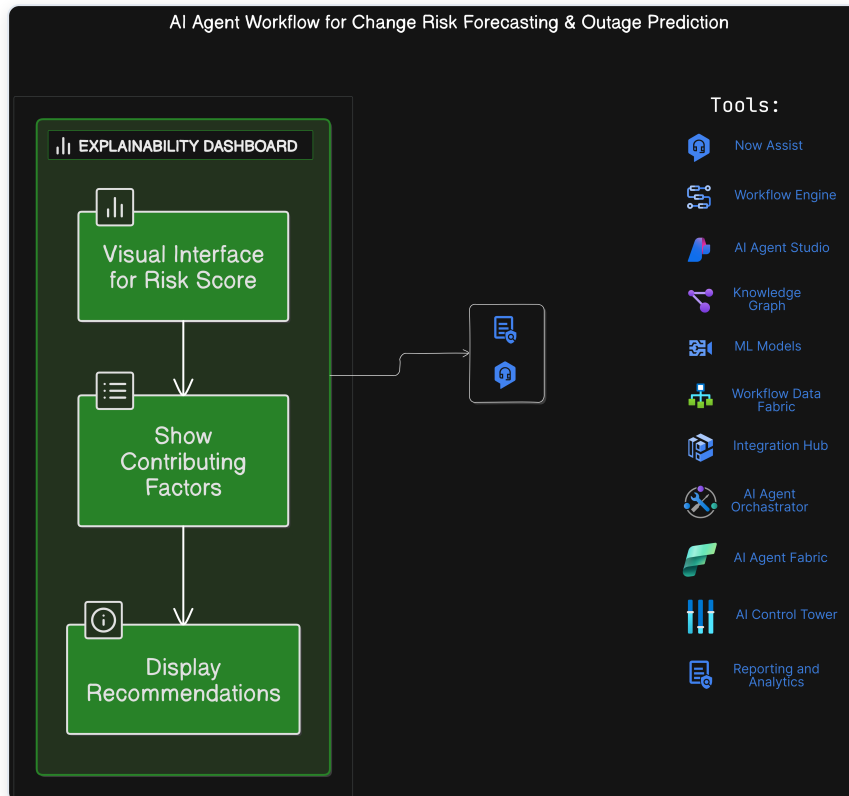
Final risk assessment with color-coded recommendations and actionable guidance

Step 6: Post-Incident Analysis



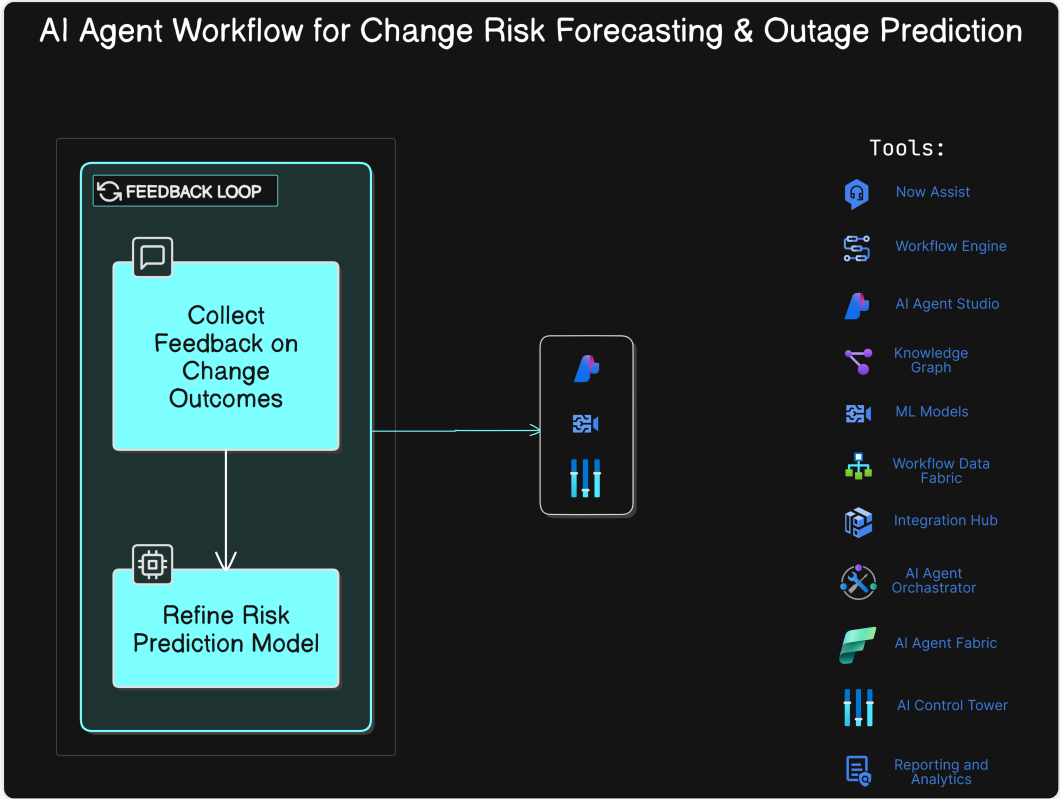
Root cause analysis and resolution recommendations for post-deployment incidents

Step 7: Explainability Dashboard



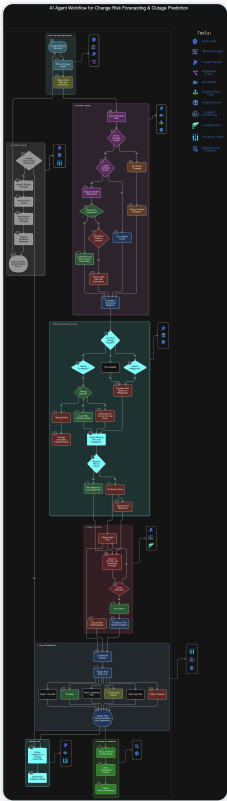
Transparent visualization of risk factors and decision-making process

Step 8: Continuous Learning Feedback Loop

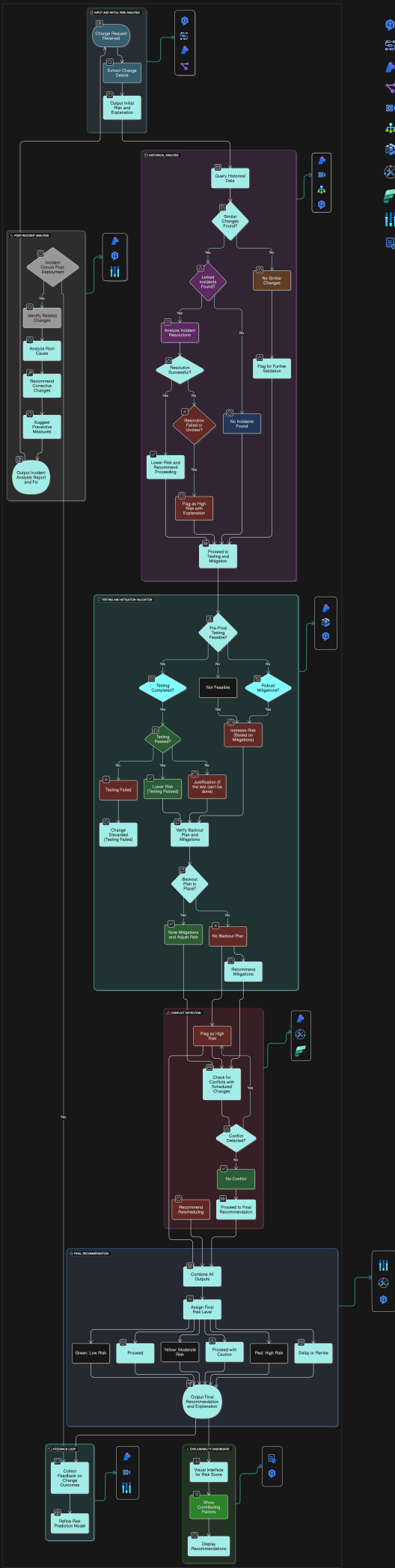


Continuous improvement through feedback collection and model refinement

Complete SRE Agent Workflow



Comprehensive overview showing the complete AI Agent workflow for change risk forecasting and outage prediction



Tools:

- Now Assist
- Workflow Engine
- AI Agent Studio
- Knowledge Graph
- ML Models
- Workflow Data Fabric
- Integration Hub
- AI Agent Orchestrator
- AI Agent Fabric
- AI Control Tower
- Reporting and Analytics